Lab2

Part1

Link for this Image in my Docker hub account:

https://hub.docker.com/repository/docker/ahmedamer10/iti-flask-lab2/general

Steps for making the image:

Step1: Install Git to Clone the Repository

sudo dnf install -y git
git clone https://github.com/meldafrawi/basic-flask-app.git
cd basic-flask-app

```
[root@server ~]# git clone https://github.com/meldafrawi/basic-flask-app.git
Cloning into 'basic-flask-app'...
remote: Enumerating objects: 32, done.
remote: Counting objects: 100% (21/21), done.
remote: Compressing objects: 100% (13/13), done.
Receiving objects: 100% (32/32), 274.91 KiB | 376.00 KiB/s, done.
remote: Total 32 (delta 8), reused 8 (delta 8), pack-reused 11 (from 1)
Resolving deltas: 100% (8/8), done.
[root@server ~]#
[root@server ~]# cd basic-flask-app
```

Step2: Create Dockerfile

```
[root@server basic-flask-app]# vim Dockerfile
[root@server basic-flask-app]# cat Dockerfile
FROM python:3.9-alpine

WORKDIR /app

COPY . .

RUN pip install --no-cache-dir flask

EXPOSE 5000

CMD ["python", "routes.py"]
[root@server basic-flask-app]#
```

```
root@server:~/basic-flask-app — vim Dockerfile

FROM python:3.9-alpine

WORKDIR /app

COPY . .

RUN pip install --no-cache-dir flask

EXPOSE 5000

CMD ["python", "routes.py"]
~
~
~
~
~
~
~
~
~
~
~
-- INSERT --                                          11,26          All
```

Step3: Build the Image

docker build -t iti-flask-lab2 .

```
[root@server basic-flask-app]# vim Dockerfile
[root@server basic-flask-app]# docker build -t iti-flask-lab2 .
[+] Building 13.3s (9/9) FINISHED                                                                                  docker:default
 => [internal] load build definition from Dockerfile                                                                          0.0s
 => => transferring dockerfile: 224B                                                                                          0.0s
 => [internal] load metadata for docker.io/library/python:3.9-alpine                                                          2.6s
 => [internal] load .dockerignore                                                                                             0.0s
 => => transferring context: 2B                                                                                               0.0s
 => [1/4] FROM docker.io/library/python:3.9-alpine@sha256:c99b6eb43b3ac4d750db3d6e8b22268d5ea9a99deead7218ce3deda7f2ca029c     0.0s
 => => resolve docker.io/library/python:3.9-alpine@sha256:c99b6eb43b3ac4d750db3d6e8b22268d5ea9a99deead7218ce3deda7f2ca029c     0.0s
 => [internal] load build context                                                                                             0.0s
 => => transferring context: 6.29kB                                                                                           0.0s
 => CACHED [2/4] WORKDIR /app                                                                                                 0.0s
 => [3/4] COPY . .                                                                                                            0.0s
 => [4/4] RUN pip install --no-cache-dir flask                                                                                9.7s
 => exporting to image                                                                                                        0.9s
 => => exporting layers                                                                                                       0.6s
 => => exporting manifest sha256:3178963dc0099428baf15f0b5a057cb5ad24eae1f6a4ff3c1fffc528b7b032d9                             0.0s
 => => exporting config sha256:761f0598d7309cecc83b87562ff17461f04499abdb6e05d466b31bda7cabce4a                               0.0s
 => => exporting attestation manifest sha256:52a710ca705bcb378319caff1c7b5a9a6ffcc1d2b3e4a77fa139be0589d0292b                 0.0s
 => => exporting manifest list sha256:d8d49ae757644ae2c47dec2858f71b8b2dd73bc6bd7f0aa4060fe61e4c4da63e                        0.0s
 => => naming to docker.io/library/iti-flask-lab2:latest                                                                      0.0s
 => => unpacking to docker.io/library/iti-flask-lab2:latest                                                                   0.3s
[root@server basic-flask-app]#
```

Step4: Run the Container with Memory Limit

docker run -d \

--name iti-flask-container \

--memory="100m" \

-p 80:5000 \

iti-flask-lab2

```
[root@server basic-flask-app]# docker run -d \
--name iti-flask-container \
--memory="100m" \
-p 80:5000 \
iti-flask-lab2
1a4d3570d915181eeaef57b37387988fc30d4e0e329b33bd3d6c9869b74c90ff
[root@server basic-flask-app]# docker ps
CONTAINER ID   IMAGE           COMMAND             CREATED         STATUS         PORTS                                   NAMES
1a4d3570d915   iti-flask-lab2  "python routes.py"  15 seconds ago  Up 14 seconds  0.0.0.0:80->5000/tcp, [::]:80->5000/tcp  iti-flask-container
```

Step 5: Push Image to Docker Hub

docker login
docker tag iti-flask-lab2 ahmedamer10/iti-flask-lab2
docker push ahmedamer10/iti-flask-lab2

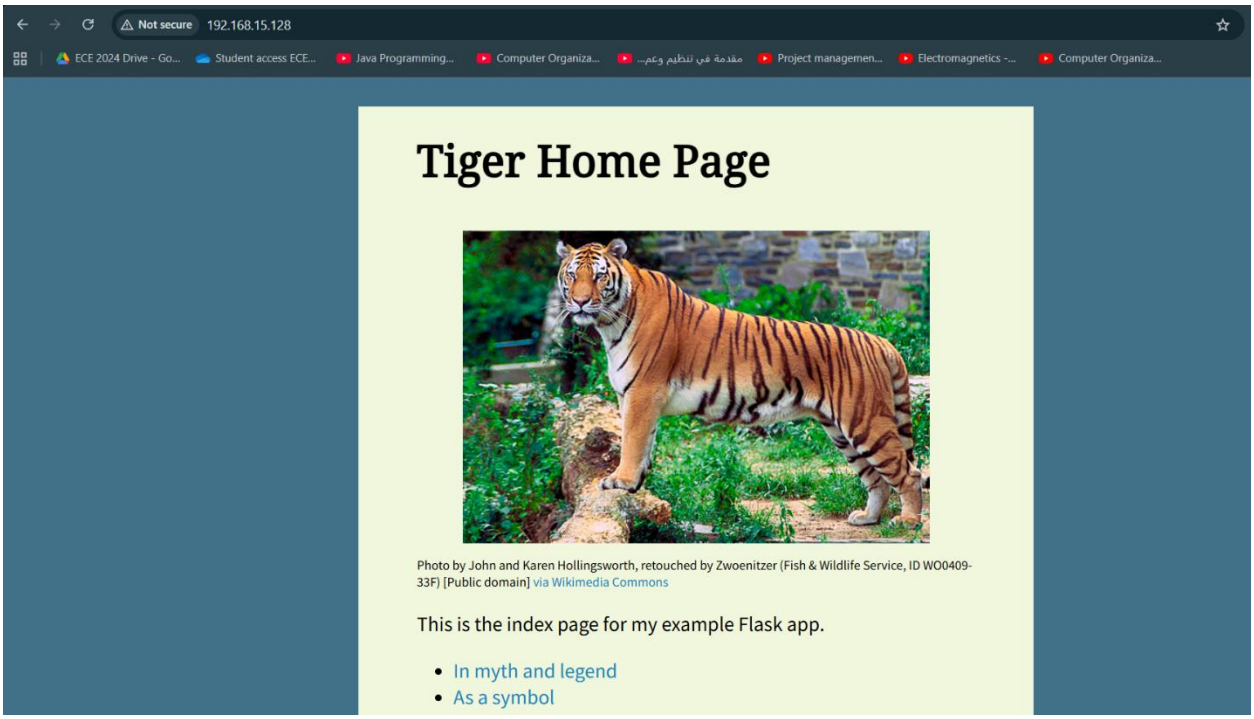
```
[root@server basic-flask-app]# docker login
Authenticating with existing credentials... [Username: ahmedamer10]

i Info → To login with a different account, run 'docker logout' followed by 'docker login'


Login Succeeded
[root@server basic-flask-app]# docker push ahmedamer10/iti-flask-lab2:latest
The push refers to repository [docker.io/ahmedamer10/iti-flask-lab2]
9806928fa0da: Pushed
2d35ebdb57d9: Pushed
c9008197b8dd: Pushed
b410233c9818: Pushed
28264e704d4d: Pushed
27711589a568: Pushed
cde5eaaf9876: Pushed
34b596050a95: Pushed
latest: digest: sha256:d8d49ae757644ae2c47dec2858f71b8b2dd73bc6bd7f0aa4060fe61e4c4da63e size: 856
[root@server basic-flask-app]#
```

Step 6: Test the Container

http://<VM-IP>



Part2

Step 1: Create Network

docker network create \
--driver bridge \
--subnet 10.0.0.0/8 \
iti-network

Step 2: Create Folder for Volume

mkdir web-content
cd web-content

```
root@server:~/web-content
[root@server ~]# docker network create \
> --driver bridge \
> --subnet 10.0.0.0/8 \
> iti-network
cb7596b9f4eb726f05157a27bded24113abdc2440a32309bc7d821f8276d21eb
[root@server ~]#
[root@server ~]# mkdir web-content
[root@server ~]# cd web-content
```

Step 3: Create index.html

vim index.html

```
root@server:~/web-content — vim index.html
<h1>Lab 2 ITI - Ahmed</h1>
~
~
~
```

Step 4: Run Nginx Container

docker run -d \
--name webserver-iti \
--network iti-network \
-p 8080:80 \
-v $(pwd)/index.html:/usr/share/nginx/html/index.html \
nginx:alpine

```
[root@server web-content]# docker run -d \
--name webserver-iti \
--network iti-network \
-p 8080:80 \
-v $(pwd)/index.html:/usr/share/nginx/html/index.html \
nginx:alpine
7c4ccf43accf8089662b936bc8b288ff65ae24c7b9f7441f9a7a5173a8e111e4
[root@server web-content]#
```

Step 5: Test

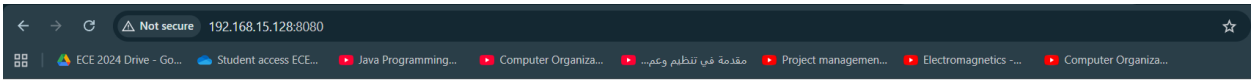
From VM

curl http://localhost:8080

```
[root@server web-content]# curl http://localhost:8080
<h1>Lab 2 ITI - Ahmed</h1>
[root@server web-content]#
```

From browser in windows

http:// <VM-IP>:8080



Lab 2 ITI - Ahmed